

Testing Strategies, Quality Assurance & DevOps CI/CD - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Software Engineering | Chapter: Chapter 3: Software Testing, Quality Assurance & DevOps CI/CD | Academic Year: 2026-2027

1. The Testing Pyramid & Testing Levels

Software testing verifies that software meets requirements and identifies defects.

The Testing Pyramid advocates for a broad base of fast, automated Unit Tests, followed by Integration Tests, and a smaller apex of End-to-End (E2E) UI Tests.

Unit Testing: Validates individual units/functions in isolation, isolating external dependencies using test doubles (mocks, stubs, spies).

Integration Testing: Validates interactions between integrated components (e.g., Spring Boot repository talking to real MySQL database).

System / E2E Testing: Validates the entire integrated system end-to-end against business specifications.

2. Black-Box versus White-Box Test Design

Black-Box Testing: Tests functionality without knowledge of internal code structure.

Techniques include Equivalence Class Partitioning (dividing inputs into valid/invalid classes) and Boundary Value Analysis (testing extremes: min, min+1, nominal, max-1, max).

White-Box Testing: Tests internal logic, data structures, and control flow. Metrics include Statement Coverage, Branch/Decision Coverage, and Cyclomatic Complexity (McCabe metric: $V(G) = E - N + 2P$).

3. Continuous Integration, Continuous Delivery & DevOps

DevOps integrates software development (Dev) and IT operations (Ops) to shorten the systems development life cycle and provide continuous delivery with high software quality.

Continuous Integration (CI): Developers merge code into a shared repository frequently; every commit triggers automated builds and unit/integration test suites.

Continuous Delivery (CD): Code is automatically tested, built, and packaged into deployable artifacts (Docker containers) ready for one-click production release.

Continuous Deployment: Every build that passes all automated test gates is automatically released to production environments without human intervention.